

BT101

Introductory Biology

(Lecture 4)

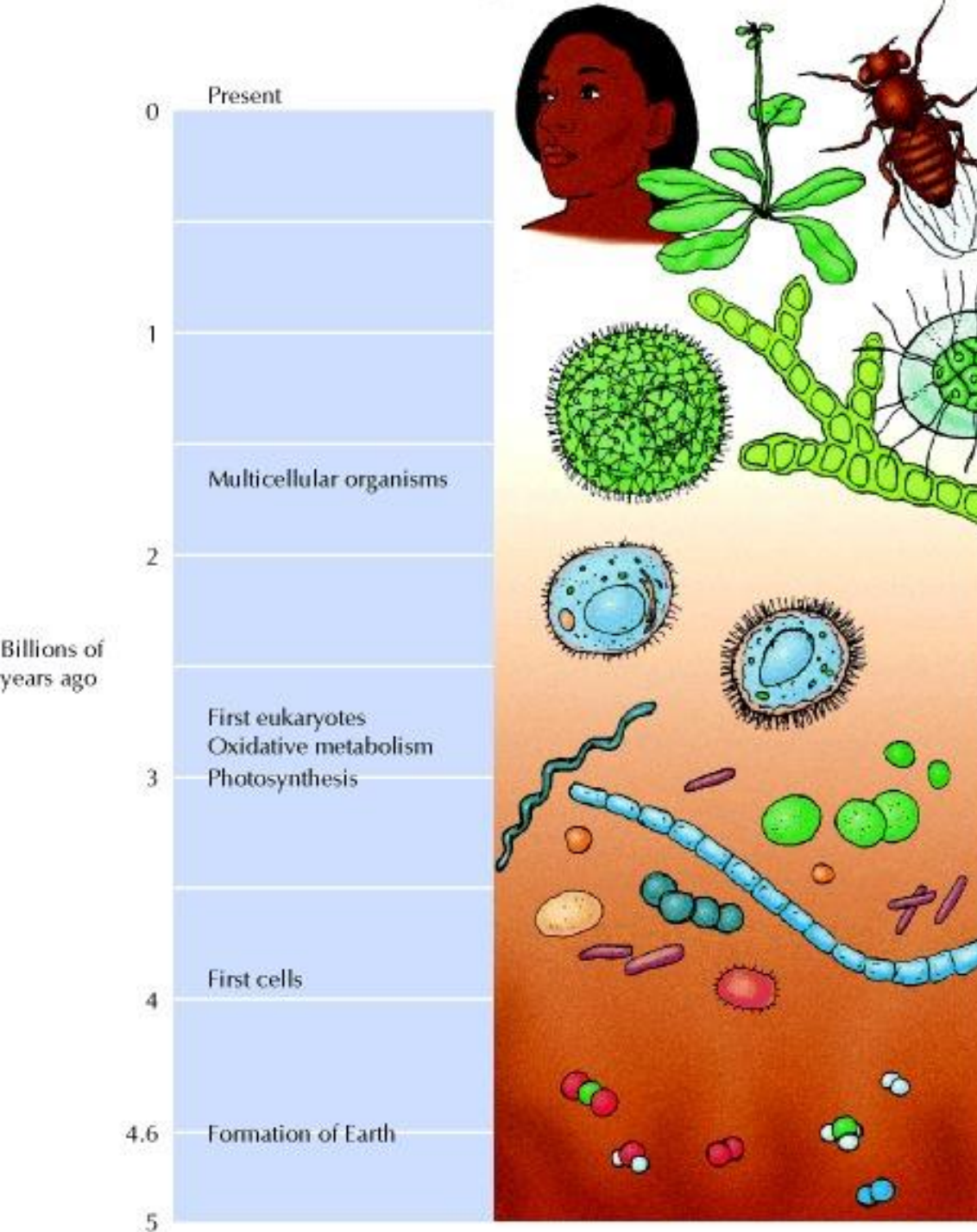
Dr. Rajkumar P Thummer
Assistant Professor
Department of Biosciences and Bioengineering
IIT Guwahati
Guwahati

Introduction



- ❑ Cells are divided into **two** main classes, initially defined by whether they contain a nucleus.
- ❑ Prokaryotic cells (bacteria) lack a nuclear envelope; eukaryotic cells have a nucleus in which the genetic material is separated from the cytoplasm.
- ❑ Prokaryotic cells are generally smaller and simpler than eukaryotic cells; in addition to the absence of a nucleus, their genomes are less complex and they do not contain cytoplasmic organelles or a cytoskeleton.
- ❑ **In spite of these differences, the same basic molecular mechanisms govern the lives of both prokaryotes and eukaryotes, indicating that all present-day cells are descended from a single primordial ancestor.**
- ❑ **How did this first cell develop?**
- ❑ **And how did the complexity and diversity exhibited by present-day cells evolve?**

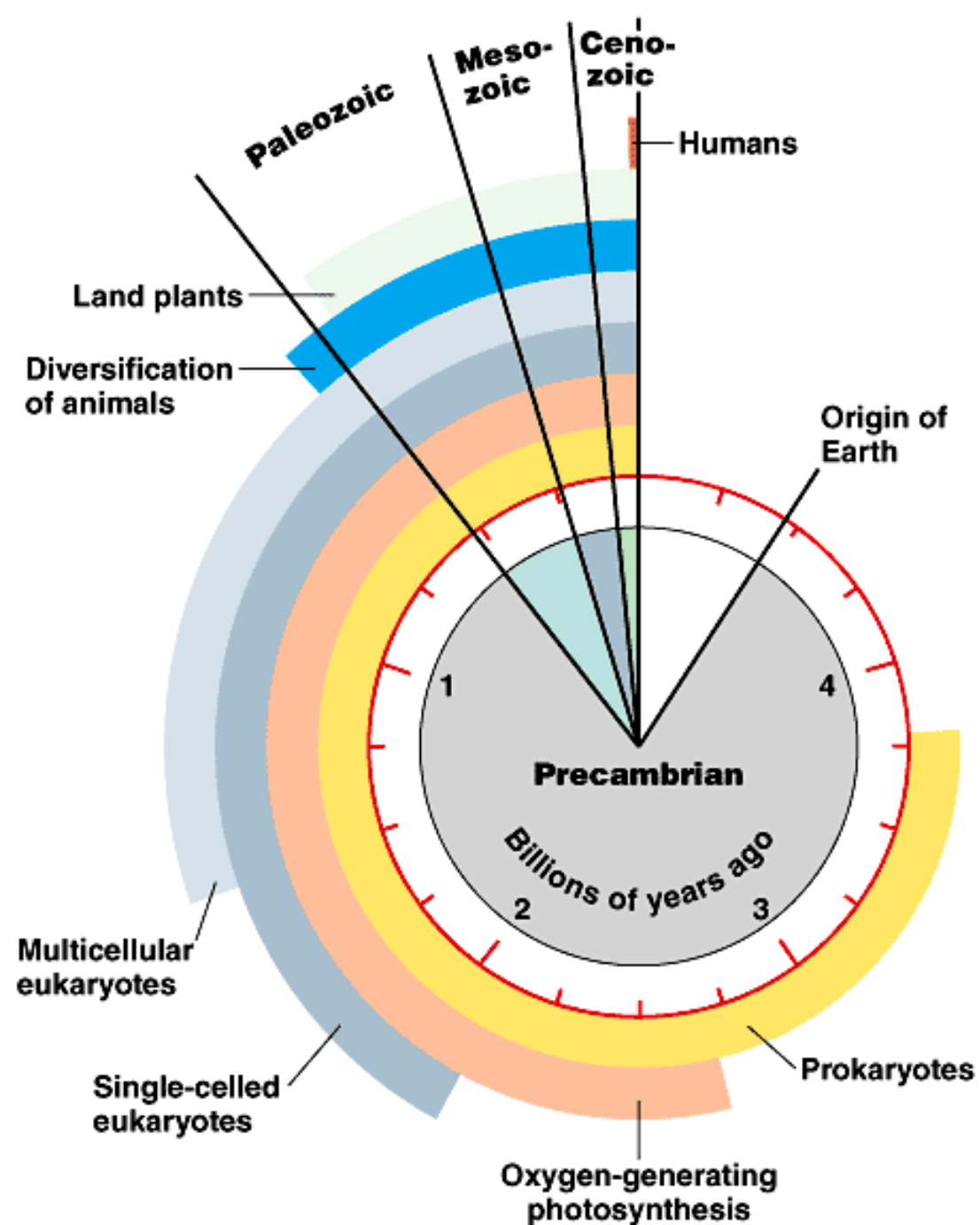
Introduction



Time scale of evolution.

The scale indicates the approximate times at which some of the major events in the evolution of cells are thought to have occurred.

Introduction



Thank you for your attention